

Degenerative Spine Disease Analysis

All of Us Research Program

Comparison of GLP-1 Agonists, Metformin, and Combination Therapy

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1. EXECUTIVE SUMMARY

This analysis examines the association between diabetes medication type (GLP-1 agonists, metformin monotherapy, or combination therapy) and incident degenerative spine disease diagnoses in adults with type 2 diabetes using data from the All of Us Research Program (n=26,350). After excluding participants with missing BMI data (n=11,225), the final analytic cohort included 15,125 individuals with mean follow-up of 3.7 years.

Finding	Result
Primary Outcome	Incident degenerative spine diagnosis within 2 years
Overall Incidence Rate	8.6% (1,301/15,125 participants)
GLP-1 vs Metformin (Main)	aOR = 1.10 (95% CI: 0.88-1.37, p=0.42) - NOT significant
Combo vs Metformin (Main)	aOR = 1.33 (95% CI: 1.08-1.64, p=0.008) - SIGNIFICANT
GLP-1×Obesity Interaction	aOR = 3.00 (95% CI: 1.41-6.37, p=0.004) - SIGNIFICANT
Key Covariate: Age	aOR = 1.03 per year (95% CI: 1.02-1.03, p<0.001)
Key Covariate: BMI	aOR = 1.01 per unit (95% CI: 1.00-1.02, p=0.005)

2. COHORT CHARACTERISTICS

2.1 Study Population and Data Source

Purpose: This section describes the baseline characteristics of the study population to establish the demographic and clinical profile of participants and assess comparability across exposure groups.

Data Source: All of Us Research Program - de-identified electronic health record (EHR) data from a diverse, nationwide U.S. cohort.

2.2 Exposure Group Distribution (N=26,350)

Medication Group	N	Percent
Metformin Only	22,741	86.3%
GLP-1 Agonist Only	1,614	6.1%
Combination Therapy	1,995	7.6%
TOTAL	26,350	100%

Interpretation: The cohort is heavily weighted toward metformin monotherapy (86.3%), reflecting standard first-line treatment for type 2 diabetes. GLP-1 agonists and combination therapy represent smaller subgroups, which is expected given their use as second-line agents or in patients with more advanced disease.

2.3 Demographic Characteristics (N=26,350)

Characteristic	Category	N	Percent
Sex at Birth	Female	15,498	58.8%
	Male	10,545	40.0%
	Other/Unknown	307	1.2%
Race	White	12,414	47.1%
	Black or African American	6,055	23.0%
	Other/Unknown	7,881	29.9%
Ethnicity	Not Hispanic or Latino	20,258	76.9%
	Hispanic or Latino	5,301	20.1%
	Unknown	791	3.0%

Interpretation: The cohort demonstrates good racial and ethnic diversity: 47.1% White, 23.0% Black/African American, and 20.1% Hispanic/Latino. The 58.8% female predominance is consistent with higher diabetes prevalence in women in some populations. This diversity is a strength of the All of Us program and enhances generalizability.

3. MISSING DATA ANALYSIS

3.1 BMI Missingness Assessment

Purpose: BMI is a critical confounder in spine disease research (obesity increases mechanical load on the spine). This analysis assesses whether BMI missingness is random or systematically related to exposure/covariates, which would introduce selection bias if data are excluded.

Method: Logistic regression model predicting BMI missingness from exposure group, demographics, and age. Robust standard errors (HC3) used to account for potential heteroscedasticity.

3.2 BMI Missingness Rates by Exposure Group

Exposure Group	Missing BMI (%)	Has BMI (%)
Metformin Only	43.7%	56.3%
GLP-1 Agonist Only	29.5%	70.5%
Combination Therapy	41.2%	58.8%

3.3 Predictors of BMI Missingness (Logistic Regression)

Predictor	Odds Ratio	95% CI	P-value	Interpretation
GLP-1 vs Combo	0.61	0.53-0.70	<0.001	GLP-1 users LESS likely to have missing BMI
Metformin vs Combo	1.10	1.00-1.21	0.041	Metformin users slightly MORE likely to have missing BMI
Female vs Male	0.75	0.71-0.79	<0.001	Females LESS likely to have missing BMI
Black vs White	0.82	0.77-0.88	<0.001	Black participants LESS likely to have missing BMI
Age (per year)	1.00	1.00-1.00	0.036	Minimal age effect

KEY FINDING: BMI missingness is NOT random - it differs significantly by medication type, sex, and race. GLP-1 users have 39% lower odds of missing BMI (likely reflecting more intensive clinical monitoring for this newer drug class). This is **Missing Not At Random (MNAR)**, which means complete-case analysis may introduce selection bias.

Implication: The subsequent analyses use only complete cases (n=15,125 with BMI data). Results should be interpreted with caution - findings may not generalize to the 43% of participants with missing BMI, who differ systematically from those with BMI data.

4. MAIN ANALYSIS: MEDICATION TYPE AND SPINE DISEASE RISK

4.1 Primary Outcome: Incident Degenerative Spine Diagnosis

Purpose: To determine whether GLP-1 agonists or combination therapy are associated with different risk of incident degenerative spine disease compared to metformin monotherapy, adjusting for known confounders.

Outcome Definition: New diagnosis of degenerative spine condition (spondylosis, disc disease, spinal stenosis, spondylolisthesis, radiculopathy, or myelopathy) occurring within 2 years after diabetes medication initiation (index date).

Analytic Sample: n=15,125 participants with complete BMI data

Statistical Model: Multivariable logistic regression with robust standard errors (HC3). Reference group: Metformin monotherapy.

Covariates: Age at index date (continuous), sex, race, ethnicity, and BMI (continuous).

4.2 Primary Analysis Results (N=15,125)

Variable	Adjusted OR	95% CI	P-value
EXPOSURE (vs Metformin):			
GLP-1 Agonist	1.10	0.88 - 1.37	0.422
Combination Therapy	1.33	1.08 - 1.64	0.008**
DEMOGRAPHICS:			
Female (vs Male)	1.11	0.97 - 1.26	0.122
Other/Unknown sex	1.29	0.77 - 2.15	0.329
Black (vs White)	1.18	1.01 - 1.37	0.036*
Other/Unknown race	1.03	0.83 - 1.27	0.786
Not Hispanic (vs Hispanic)	0.99	0.79 - 1.25	0.953
Unknown ethnicity	1.49	1.06 - 2.10	0.023*
CONTINUOUS COVARIATES:			
Age (per year)	1.03	1.02 - 1.03	<0.001***
BMI (per kg/m²)	1.01	1.00 - 1.02	0.005**

* p<0.05, ** p<0.01, *** p<0.001

KEY FINDINGS:

1. GLP-1 Agonists vs Metformin: NO significant association (aOR=1.10, p=0.422). GLP-1 users have similar spine disease risk as metformin users after adjusting for confounders.

2. Combination Therapy vs Metformin: SIGNIFICANT 33% increased odds (aOR=1.33, p=0.008).

Patients on combination therapy have higher spine disease risk. This likely reflects *confounding by indication* - combination therapy is prescribed to patients with more advanced diabetes, who may have longer disease duration, higher cumulative glycemic burden, and more comorbidities (all of which increase spine disease risk).

3. Age Effect: Strong predictor - each additional year increases odds by 3% (aOR=1.03, $p<0.001$). A 60-year-old has ~2.4× the odds of a 30-year-old, consistent with degenerative nature of spine disease.

4. BMI Effect: Modest but significant - each 1 kg/m² increase in BMI increases odds by 1% (aOR=1.01, $p=0.005$). A person with BMI 35 has ~1.22× the odds of BMI 25, reflecting mechanical stress from excess weight.

5. Black Race: 18% increased odds vs White (aOR=1.18, $p=0.036$). May reflect health disparities in diabetes management, access to care, or structural/occupational risk factors.

5. INTERACTION ANALYSIS: MEDICATION × OBESITY

5.1 Rationale for Interaction Model

Purpose: To test whether the effect of medication type on spine disease risk differs by obesity status (BMI ≥ 30 vs < 30). This addresses the hypothesis that GLP-1 agonists, which promote weight loss, may be particularly protective in obese individuals.

Method: Logistic regression including medication $\times$ obesity interaction terms. The model compares four groups: (1) Metformin + Non-obese [reference], (2) Metformin + Obese, (3) GLP-1 + Non-obese, (4) GLP-1 + Obese, plus similar strata for combination therapy.

Statistical Test: Joint Wald test evaluates whether BOTH interaction terms (GLP-1 $\times$ obesity AND combo $\times$ obesity) are simultaneously zero. Rejection indicates effect modification.

5.2 Interaction Model Results

Term	Adjusted OR	95% CI	P-value
MAIN EFFECTS:			
GLP-1 (in non-obese)	0.43	0.21 - 0.88	0.021*
Combo (in non-obese)	1.29	0.71 - 2.32	0.405
Obesity (on metformin)	0.96	0.80 - 1.14	0.623
INTERACTION TERMS:			
GLP-1 $\times$ Obesity	3.00	1.41 - 6.37	0.004**
Combo $\times$ Obesity	1.29	0.63 - 2.32	0.772
Joint Wald Test:	$\chi^2=8.65$	df=2	p=0.013*

* p<0.05, ** p<0.01

5.3 Interpretation: Effect Modification by Obesity

CRITICAL FINDING: Significant interaction between GLP-1 agonists and obesity (p=0.004). The effect of GLP-1 on spine disease risk *differs substantially* depending on obesity status:

In NON-OBESE patients (BMI < 30):

- GLP-1 vs Metformin: aOR = 0.43 (95% CI: 0.21-0.88, p=0.021)
- **Interpretation:** GLP-1 users have 57% LOWER odds of spine disease compared to metformin users. This is PROTECTIVE.

In OBESE patients (BMI ≥ 30):

- GLP-1 vs Metformin: aOR = $0.43 \times 3.00 = 1.29$
- **Interpretation:** The protective effect is completely reversed. In obese patients, GLP-1 users have 29% HIGHER odds than metformin users (though not statistically significant in this stratum alone).

Why might this occur?

Hypothesis 1 (Confounding by indication): Obese patients prescribed GLP-1s may have more severe/longer-standing diabetes, higher inflammation, or more comorbidities that increase spine risk. The medication choice reflects disease severity rather than drug effect.

Hypothesis 2 (Detection bias): GLP-1 users may have more frequent healthcare encounters, leading to increased diagnosis of spine conditions that would go undetected in metformin users.

Hypothesis 3 (True biological interaction): GLP-1 mechanisms (incretin effects, inflammation modulation) may operate differently in obese vs lean patients.

Clinical Implication: This finding is UNEXPECTED and requires cautious interpretation. It does NOT support a protective effect of GLP-1s in obese diabetics for spine disease prevention.

6. SENSITIVITY ANALYSIS: FOLLOW-UP TIME ADJUSTMENT

6.1 Purpose of Sensitivity Analysis

Rationale: The primary analysis does not account for variable follow-up time across participants. If follow-up duration differs systematically by medication type (e.g., GLP-1 users have shorter observation periods), this could bias the outcome ascertainment.

Method: Re-run the main model adding log(follow-up days) as a covariate. If the medication effects remain stable, this supports robustness to follow-up time differences.

6.2 Follow-Up Adjusted Model Results

Variable	Original aOR	Follow-Up Adjusted aOR	Change
GLP-1 vs Metformin	1.10 (0.88-1.37)	1.11 (0.88-1.39)	Stable
Combo vs Metformin	1.33 (1.08-1.64)**	1.33 (1.08-1.64)**	Stable
Age (per year)	1.03 (1.02-1.03)***	1.03 (1.02-1.03)***	Stable
BMI (per unit)	1.01 (1.00-1.02)**	1.01 (1.00-1.02)**	Stable
Log(follow-up days)	—	1.02 (0.91-1.15)	p=0.677 (NS)

Conclusion: Follow-up time is NOT a significant predictor of spine disease diagnosis (aOR=1.02, p=0.677), and adding it to the model does not change any of the primary findings. This suggests that differential follow-up across medication groups is not biasing the results. The main analysis findings are **robust**.

7. OVERALL CONCLUSIONS AND CLINICAL IMPLICATIONS

7.1 Summary of Key Findings

1. **GLP-1 Agonists vs Metformin (Main Effect):** No significant difference in spine disease risk in the overall population (aOR=1.10, p=0.422).
2. **Combination Therapy vs Metformin:** Significantly increased risk (aOR=1.33, p=0.008), likely reflecting confounding by indication (sicker patients receive combination therapy).
3. **GLP-1 × Obesity Interaction:** Strong and unexpected effect modification. GLP-1s appear protective in non-obese patients (aOR=0.43) but not in obese patients (aOR=1.29), with statistically significant interaction (p=0.004).
4. **Age and BMI:** Strong independent risk factors, as expected for degenerative spine disease.
5. **Racial Disparities:** Black race associated with 18% increased risk (p=0.036), highlighting potential health equity issues.

7.2 Limitations

- **Missing Data:** 43% of participants excluded due to missing BMI, which is not random (MNAR). Results may not generalize to those with missing data.
- **Confounding by Indication:** Medication choice is not randomized - patients on GLP-1s or combination therapy likely differ from metformin users in unmeasured ways (disease severity, duration, comorbidities).
- **Short Follow-Up:** Median ~3.7 years may not capture long-term spine disease development.
- **Outcome Ascertainment:** Relies on EHR diagnoses, which may reflect healthcare utilization patterns rather than true disease incidence (detection bias).
- **Residual Confounding:** Lack of data on physical activity, occupation, smoking, prior spine injury, and other important spine disease risk factors.

7.3 Clinical Implications

- **GLP-1s and Spine Health:** This analysis does NOT support the hypothesis that GLP-1 agonists reduce spine disease risk through weight loss. The unexpected interaction with obesity requires further investigation before clinical recommendations can be made.
- **Patient Selection:** Clinicians should continue selecting diabetes medications based on glycemic control, cardiovascular benefits, and patient preference. Spine disease prevention is not currently an indication for GLP-1 therapy.
- **Health Equity:** Observed racial disparities warrant targeted interventions to improve diabetes management and spine health in Black patients.

7.4 Recommendations for Future Research

- Conduct propensity score matching or inverse probability weighting to better account for confounding by indication
- Stratify analyses by diabetes duration and severity to isolate medication effects from disease progression
- Include imaging-based spine outcomes (e.g., disc height, stenosis severity) to reduce detection bias
- Extend follow-up to 5-10 years to capture long-term degenerative processes
- Investigate mechanisms underlying the GLP-1×obesity interaction through mediation analysis (weight change, inflammation markers, glycemic control)

8. TECHNICAL APPENDIX

8.1 Statistical Methods

Software: Python with statsmodels library (version $\geq 0.14.0$)

Regression Type: Binary logistic regression (outcome: incident spine disease, yes/no)

Standard Errors: Heteroscedasticity-consistent (HC3) robust standard errors to account for variance heterogeneity across covariate patterns. HC3 is preferred for finite samples and influential observations.

Convergence: All models converged using maximum likelihood estimation (MLE). Newton method used for optimization.

Model Fit: Pseudo R^2 (McFadden) = 0.017 for main model, indicating that observed covariates explain ~1.7% of variance in spine disease risk. Low R^2 is typical in health outcomes with many unmeasured determinants.

Collinearity: Not formally assessed with VIF, but BMI and obesity indicator are not included in same model to avoid perfect collinearity.

Sample Size: With $n=15,125$ and 1,301 events (8.6% incidence), the analysis has >80% power to detect ORs ≥ 1.25 at $\alpha=0.05$.

8.2 Variable Definitions

Variable	Definition	Coding
Exposure	Diabetes medication type at index date	Metformin/GLP-1/Combo
Outcome	Incident degenerative spine diagnosis within 2 years	Binary (yes/no)
BMI	Body mass index (kg/m^2)	Continuous
Obesity	$\text{BMI} \geq 30 \text{ kg}/\text{m}^2$	Binary (yes/no)
Age	Age at index date (years)	Continuous
Sex	Sex at birth	Female/Male/Other
Race	Self-reported race	White/Black/Other/Unknown
Ethnicity	Hispanic ethnicity	Yes/No/Unknown
Follow-up	Days from index to last observation	Continuous (log-transformed)

8.3 Data Source

Database: All of Us Research Program

Version: Controlled Tier Dataset (access date: February 2026)

IRB Approval: All of Us Institutional Review Board (participants provided broad consent for secondary

research use)

Data Standardization: OMOP (Observational Medical Outcomes Partnership) Common Data Model

ICD-10 Codes for Degenerative Spine Disease: M47.xx (spondylosis), M48.0x (spinal stenosis), M50.3x/M51.3x (disc degeneration), M43.1x (spondylolisthesis), M54.1x (radiculopathy)

END OF REPORT

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Degenerative Spine Disease and Diabetes Medications

All of Us Research Program Analysis

Comparing GLP-1 Agonists, Metformin, and Combination Therapy

Report Generated: February 13, 2026

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DEGENERATIVE SPINE CONDITIONS INCLUDED IN ANALYSIS

This analysis identified incident degenerative spine disease diagnoses using standardized SNOMED CT concept codes from the All of Us OMOP (Observational Medical Outcomes Partnership) Common Data Model. All participants were required to have NO pre-existing spine diagnosis before their diabetes medication index date to ensure only incident (new-onset) cases were captured.

The following table lists all degenerative spine conditions included in the analysis:

SNOMED Concept ID	Condition Name	Anatomic Location	Description
137548	Spondylosis	Multiple levels	Spinal osteoarthritis (wear and tear)
198520	Spinal stenosis	Multiple levels	Narrowing of spinal canal
608177	Degenerative disc disease	Multiple levels	Age-related disc degeneration
761918	Spondylolisthesis	Vertebral column	Vertebra slippage over another
80816	Spondylosis without myelopathy	Multiple levels	Osteoarthritis, no spinal cord compression
80813	Lumbosacral spondylosis	Lumbosacral junction	Lower back osteoarthritis
77079	Cervical spondylosis	Cervical spine	Neck osteoarthritis
79119	Thoracic spondylosis	Thoracic spine	Mid-back osteoarthritis
4187244	Lumbar spinal stenosis	Lumbar spine	Lower back canal narrowing
4167097	Cervical spinal stenosis	Cervical spine	Neck canal narrowing
4134121	Lumbar radiculopathy	Lumbar nerve roots	Lower back nerve compression (sciatica)
4134122	Cervical radiculopathy	Cervical nerve roots	Neck nerve compression (arm pain)
36717608	Cervical myelopathy	Cervical spinal cord	Spinal cord dysfunction in neck

Note on Selection Criteria: These conditions represent true age-related degenerative pathology rather than congenital abnormalities, acute trauma, inflammatory diseases (e.g., ankylosing spondylitis), or infectious processes. SNOMED CT codes were selected using the All of Us Concept Set Builder with hierarchical descendant mapping to capture all relevant diagnostic subtypes within each category.

Data Source: Diagnoses extracted from *condition_occurrence* table in OMOP CDM, which aggregates ICD-10-CM codes from multiple healthcare systems and maps them to standardized SNOMED concepts for research use.

SCIENTIFIC APPROACH AND METHODOLOGY

1. Study Design

Type: Retrospective cohort study

Data Source: All of Us Research Program Registered Tier Dataset v8 (R2024Q3R8)

Population: Adults aged 40-65 with type 2 diabetes mellitus

Sample Size: 26,350 (baseline cohort) → 15,125 (complete-case analysis with BMI)

Follow-up: Mean 3.7 years (minimum 2 years required)

2. Temporal Design and Index Date

Critical Methodological Feature: The "index date" for each participant was defined as the date of first exposure to their diabetes medication (metformin and/or GLP-1 agonist). This establishes a clear temporal sequence:

Timeline: Index Date (medication start) → Follow-up period (2 years) → Outcome assessment (spine diagnosis yes/no)

This design ensures that:

- Exposure precedes outcome (temporal precedence for causality)
- Only NEW spine diagnoses after medication initiation are counted (incident cases)
- Follow-up time is standardized across participants

Exclusion of Prevalent Cases: Any participant with a degenerative spine diagnosis AT or BEFORE their index date was excluded from outcome analysis. This is essential to avoid reverse causation (where existing spine disease might influence medication choice).

3. Exposure Classification

Participants were classified into three mutually exclusive groups based on medication dispensing records in the *drug_exposure* table within 90 days of index date:

1. Metformin Monotherapy (n=22,741, 86%): Metformin without GLP-1 agonist

- Represents standard first-line diabetes treatment per ADA guidelines
- Longest clinical experience, lowest cost, established safety profile

2. GLP-1 Agonist Monotherapy (n=1,614, 6%): GLP-1 agonist without metformin

- Includes semaglutide, liraglutide, dulaglutide, exenatide (SNOMED concept 1503297)
- Newer agents with weight loss and cardiovascular benefits
- Typically prescribed when metformin contraindicated or for patients prioritizing weight loss

3. Combination Therapy (n=1,995, 8%): Both metformin AND GLP-1 agonist

- Represents intensified therapy for inadequate glycemic control
- Patients likely have more severe or longer-standing diabetes
- Highest risk of confounding by indication

4. Outcome Definition

Primary Outcome: Incident degenerative spine disease diagnosis within 2 years post-index

Operationalization: Binary variable (yes/no) indicating whether the participant received ANY of the 13 qualifying spine diagnoses (see previous table) after their index date with:

- No prior diagnosis of that specific condition before index date
- Minimum 730 days follow-up (to ensure adequate observation period)

Rationale for 2-Year Window:

- Balances medium-term drug effect capture vs. patient attrition
- Reduces confounding from medication switching (common after 2-3 years)
- Aligns with typical RCT follow-up periods for diabetes medications

Limitation Acknowledged: Outcome depends on healthcare utilization (detection bias). Patients on newer medications (GLP-1s) may have more frequent clinic visits, leading to increased diagnosis of asymptomatic spine conditions.

5. Confounding Control Strategy

The Central Challenge: In observational studies, medication choice is not random. It reflects clinical decision-making based on patient characteristics we can measure (age, BMI) and many we cannot (diabetes duration, HbA1c, comorbidities, socioeconomic status, occupation).

Measured Confounders (Adjusted in Models):

- **Age:** Continuous (years) - strongest predictor of degenerative spine disease
- **BMI:** Continuous (kg/m^2) - obesity increases mechanical spine load
- **Sex:** Categorical (Male, Female, Other/Unknown) - biomechanical differences
- **Race:** Categorical (White, Black/African American, Other/Unknown) - health disparities
- **Ethnicity:** Categorical (Hispanic/Latino, Not Hispanic/Latino, Unknown)

Unmeasured Confounders (Acknowledged Limitations):

- Diabetes duration and severity (HbA1c, fasting glucose)
- Comorbidity burden (Charlson index, prior cardiovascular events)
- Physical activity and occupation (manual labor increases spine risk)
- Smoking status (incomplete data, excluded from models)
- Prior spine injury or structural abnormalities

Statistical Approach: Multivariable logistic regression with heteroscedasticity-robust (HC3) standard errors. Reference categories explicitly specified to avoid interpretation ambiguity.

6. Missing Data Handling

BMI Missingness: 42.6% of participants (11,225/26,350) lacked recorded BMI measurements.

Our Approach (Transparent and Conservative):

1. Assessed missingness pattern using logistic regression (Section 3)

2. **Found:** Missingness NOT random - differed by medication type, sex, race (MNAR)
3. **Decision:** Complete-case analysis (exclude missing BMI) rather than imputation
4. **Justification:** Multiple imputation requires MAR assumption, which was violated

Implication: Results may not generalize to the 43% with missing BMI. We explicitly report the missingness model to allow readers to judge potential selection bias.

7. Effect Modification (Interaction) Analysis

Hypothesis: The effect of GLP-1 agonists on spine disease might differ by obesity status, since GLP-1s promote weight loss which could reduce mechanical spine load.

Test: Included medication×obesity ($BMI \geq 30$) interaction terms in logistic regression. Joint Wald test assessed whether both interaction coefficients were simultaneously zero.

Interpretation if Significant: The medication-spine association differs for obese vs. non-obese patients, requiring stratified results reporting.

8. Software and Reproducibility

Platform: All of Us Researcher Workbench (cloud Jupyter notebooks)

Language: Python 3.x

Libraries: statsmodels (≥ 0.14), pandas, numpy

Data Model: OMOP Common Data Model v5.3

Dataset: All of Us Registered Tier v8 (R2024Q3R8)

IRB: All of Us Institutional Review Board (participants provided broad consent)

EXECUTIVE SUMMARY

This analysis examines the association between diabetes medication type and incident degenerative spine disease in 15,125 adults with type 2 diabetes from the All of Us Research Program (mean follow-up: 3.7 years, outcome incidence: 8.6%).

Finding	Result
Primary Outcome	Incident degenerative spine diagnosis within 2 years
Overall Incidence	8.6% (1,301/15,125)
GLP-1 vs Metformin	aOR=1.10 (0.88-1.37), p=0.42 - NOT significant
Combo vs Metformin	aOR=1.33 (1.08-1.64), p=0.008 - SIGNIFICANT
GLP-1×Obesity Interaction	aOR=3.00 (1.41-6.37), p=0.004 - SIGNIFICANT
Age Effect	aOR=1.03 per year, p<0.001
BMI Effect	aOR=1.01 per kg/m², p=0.005

KEY TAKEAWAYS:

- 1. GLP-1 agonists show NO overall effect on spine disease risk vs. metformin
- 2. Combination therapy users have 33% higher risk (likely confounding by indication)
- 3. UNEXPECTED finding: GLP-1×obesity interaction suggests protective effect in non-obese patients (aOR=0.43) but not in obese patients - requires cautious interpretation
- 4. Age and BMI are strong independent risk factors as expected

[The complete report continues with detailed sections on:]

- Cohort Characteristics (demographics, exposure distribution)
- Missing Data Analysis (BMI missingness assessment)
- Main Regression Models (detailed results tables)
- Interaction Analysis (medication × obesity)
- Sensitivity Analyses (follow-up adjustment)
- Clinical Implications and Limitations
- Technical Appendix (variable definitions, statistical methods)

For space efficiency, full tables and extended discussions are in the companion document.